Definition (Permutation category)
The *permutation category* **Pm** is

1. *Objects*: the natural numbers $\mathbb{N}$;
2. *Morphisms*: permutations in $\mathbf{Perms}(n)$, for all $n \in \mathbb{N}$.
3. *Composition*: composition of permutations;
4. *Identities*: identity functions (which are permutations). We will denote the identity permutation in $\mathbf{Perms}(n)$ by 1_n.